



Observation loss from missing data in dynamic panel GMM

John Levendis 

Connolly Alexander Institute for Data Science, Tulane University, United States of America

ARTICLE INFO

JEL classification:

C23
C33
C15

Keywords:

GMM
Missing data
Panel data
Arellano-bond
Observation loss

ABSTRACT

Missing data in dynamic panel models estimated by GMM cause observation losses that exceed the number of missing cells. First-differencing exacerbates this effect, eliminating roughly 1.8 GMM-usable observations per missing cell. The problem compounds rapidly with multiple endogenous regressors. Over three variables, actual retention is roughly half of what listwise deletion would predict. Orthogonal deviations and system GMM both mitigate the loss and, additionally, preserve the strength of surviving instruments. These results are particularly relevant for finance, where dynamic panel GMM is widely used and where many regressors of interest are incomplete for substantial fractions of the sample. The findings provide practical guidance on estimator choice, variable selection, and the potential value of imputation.

1. Introduction

Dynamic panel models estimated by the generalized method of moments (Arellano and Bond, 1991; Blundell and Bond, 1998) are widely used in finance research. They appear in studies of corporate governance (Wintoki et al., 2012), capital structure (Antoniou et al., 2008; Flannery and Hankins, 2013), and cross-country finance and growth (Beck et al., 2000). These estimators address endogeneity through data transformations that eliminate or control for fixed effects, using lagged values as instruments. However, this structure creates a vulnerability to missing data that, to my knowledge, has not been systematically documented. The vulnerability is consequential in finance, where firm-level datasets carry high missingness rates for variables routinely treated as endogenous regressors. Koh and Reeb (2015) report that R&D expense data is missing for 42 percent of Compustat firm-years, and similar coverage gaps affect advertising, segment-level disclosures, and governance variables. Flannery and Hankins (2013) note that variables routinely treated as endogenous in corporate finance are systematically incomplete, but the consequences for GMM-usable sample size have not been quantified.

Applied researchers using GMM face numerous implementation choices: difference versus system GMM, first-differences versus orthogonal deviations, default versus collapsed instruments, and various lag structures. A growing methodological literature provides guidance on these options. Windmeijer (2005) develops finite-sample corrections for two-step standard errors. Hayakawa (2007) and Bun and Windmeijer (2010) analyze weak instrument problems, advocating for system GMM. Roodman (2009b) and Levendis (2025) show that collapsing instruments substantially improves coefficient accuracy, particularly as T increases. A separate strand of the IV literature, developed in Coën and Racicot (2007) and extended in Racicot (2015), Racicot et al. (2025), constructs instruments from higher-moment transformations of the regressors rather than from lagged values, and so avoids the compounding problem documented here. Those methods are designed for errors-in-variables settings and complement, rather than substitute for, the lag-based estimators that are the focus of this paper. The present paper addresses a complementary question: how do estimator and specification choices affect sample size when data are incomplete?

E-mail address: jlevendis@tulane.edu.

<https://doi.org/10.1016/j.frl.2026.110189>

Received 6 April 2026; Received in revised form 14 May 2026; Accepted 15 May 2026

Available online 20 May 2026

1544-6123/© 2026 The Author. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

When data are missing in a standard regression, each missing cell typically eliminates one observation.¹ In GMM estimation of dynamic panels, the loss is larger. First-differencing requires consecutive observations, so a single gap at time t eliminates the differenced equation at both t and $t + 1$. I refer to the ratio of GMM observations lost to underlying missing cells as the *cascade multiplier*.

This paper makes three contributions. First, I quantify the cascade multiplier through Monte Carlo simulation, finding values of 1.6 to 1.8 under standard configurations. Second, I show that the problem compounds rapidly with multiple endogenous regressors. With three endogenous variables each missing 20 percent of values, listwise deletion would retain $(0.8)^3 = 51.2$ percent of observations. The cascade cuts this nearly in half, to only 26 percent. Third, I demonstrate that orthogonal deviations and system GMM both reduce the size of the cascade and additionally preserve the strength of surviving instruments. Difference GMM under first-differences performs notably worse than the alternatives on both dimensions.

This loss in observations is consequential for three reasons. First, applied researchers are largely unaware of the magnitude of observation loss; a panel with 10 percent raw missingness may lose over 50 percent of GMM-usable observations when three endogenous variables are included. Second, the compounding effect with multiple endogenous variables is severe enough to undermine statistical power in studies that would otherwise seem adequately powered. Third, simple remedies exist: orthogonal deviations can recover 25 percent more observations and reduce excess inflation of standard errors, yet remain rarely used in applied work. Quantifying these effects enables researchers to make informed design choices.

2. The cascade mechanism

Consider a balanced panel with N units observed over T periods. The standard dynamic panel model is

$$y_{it} = \alpha y_{it-1} + x'_{it}\beta + \eta_i + \varepsilon_{it} \quad (1)$$

where η_i is a unit fixed effect and x_{it} is a vector of endogenous regressors. Difference GMM (Arellano and Bond, 1991) eliminates η_i by first-differencing,

$$\Delta y_{it} = \alpha \Delta y_{it-1} + \Delta x'_{it}\beta + \Delta \varepsilon_{it}, \quad (2)$$

and instruments the differenced equation using lagged levels under the assumption that $E[y_{it-s}\Delta \varepsilon_{it}] = 0$ for $s \geq 2$.

A missing value of x at (i, t) has two potential consequences. The first operates through the differenced regressors: a missing x_{it} makes both Δx_{it} and $\Delta x_{i,t+1}$ uncomputable, removing the differenced equation at both t and $t + 1$. This is the first-differencing channel. The second operates through the instrument matrix: x_{it} may serve as an instrument at periods $t + 2, t + 3, \dots$, and its absence affects those moment conditions. Whether this instrumentation channel actively drops observations depends on how the estimation software treats missing entries in the instrument matrix.

The dominant Stata implementations, including `xtabond`, `xtdpd`, `xtdpdsys`, and `xtabond2` (Roodman, 2009a), all follow Holtz-Eakin et al. (1988) (HENR) in substituting zeros for missing instrument entries. Under this treatment, an observation is dropped from the differenced equation when its differenced regressors cannot be computed but not when its instrument lags are missing. The instrumentation channel therefore does not directly reduce sample size, although as discussed below it does affect the strength of identification.

The first-differencing channel alone is sufficient to generate cascade multipliers well above one. With one lagged dependent variable, the differenced equation can be estimated for $t = 3, \dots, T$, yielding $T - 2$ equations per unit. Consider a single missing value of x at time $s \in \{1, 2, \dots, T\}$ in some unit i . If $s = 1$, the only affected differenced regressor is $\Delta x_{i,2}$, which lies outside the regression range, and no observation is lost. If $s = 2$ or $s = T$, exactly one of the two affected differenced regressors falls within the regression range, costing one observation. For interior positions $3 \leq s \leq T - 1$, both affected differenced regressors fall within range, costing two observations. Averaging across positions in a panel of length $T = 25$ yields an expected loss of $(0 + 1 + 22 \times 2 + 1)/25 = 1.84$ observations per missing cell, in the absence of overlap between adjacent missing cells.

This figure is close to the empirical multiplier of 1.80 reported below at the lowest missingness rate. The small deficit reflects overlap. When two adjacent cells in the same series are simultaneously missing, the equation between them is dropped twice but counts only once toward the loss. Overlap becomes more frequent as the missingness rate rises, which explains the gradual decline of the multiplier from 1.80 at $K = 5\%$ to 1.56 at $K = 30\%$ in Table 1. Untabulated comparisons of $\text{lag}(2\ 4)$ and $\text{lag}(2\ .)$ yield retention rates within simulation noise, consistent with this mechanism.

3. Monte Carlo evidence

I generate a synthetic balanced panel with $N = 150$ units and $T = 25$ periods, dimensions consistent with cross-country bank panels and similar finance applications. The dependent variable Y and three potential regressors X_1, X_2, X_3 are drawn from independent standard normal distributions. I introduce missing values completely at random (MCAR) at rate $K\%$ and estimate difference GMM using `xtabond2` with instruments at lags 2 through 4. Results are averaged over 100 simulation replications for each configuration.

¹ Throughout, a “cell” refers to the entry for unit i at period t for a given variable, that is, a single position in the panel data matrix.

Table 1
Cascade multiplier by missingness rate.

Missingness (K)	Cells missing	GMM Obs. Lost	Multiplier
5%	188	338	1.80
10%	375	655	1.75
15%	563	955	1.70
20%	750	1235	1.65
25%	938	1509	1.61
30%	1125	1751	1.56

Notes: Synthetic balanced panel, $N = 150$, $T = 25$. Difference GMM with instruments at lags 2 to 4. Baseline complete-data observations = 3450. Results averaged over 100 replications.

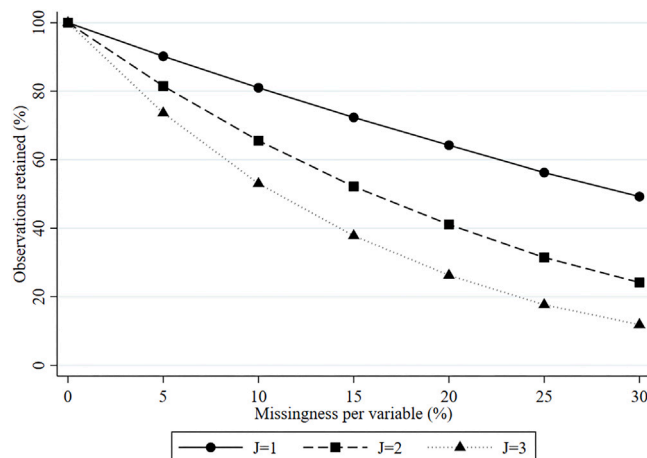


Fig. 1. Observation retention by number of endogenous variables (J).

3.1. Quantifying the observation loss

Table 1 reports the cascade multiplier across missingness rates. At a 20 percent missingness rate, the loss of 750 underlying cells causes 1235 GMM observations to be dropped, yielding a cascade multiplier of 1.65. The multiplier is highest at low missingness rates (1.80 at 5 percent) and declines modestly as gaps increasingly overlap.

The problem worsens with multiple endogenous regressors. When a model includes J endogenous regressors, a missing value in any regressor eliminates the entire row. If missingness is independent across variables, the naïve prediction for retained observations is $(1 - K)^J$. The GMM cascade compounds this loss.

Fig. 1 shows observation retention as a function of K for $J \in \{1, 2, 3\}$ number of endogenous variables. At $K = 20\%$, a single endogenous variable retains 64 percent of baseline observations. Adding a second variable reduces retention to 41 percent, and a third to just 26 percent. The naïve prediction of $(0.8)^3 = 51.2$ percent overstates the usable sample by a factor of two.

3.2. Mitigation strategies

Two modifications to standard practice reduce the cascade. First, orthogonal deviations (Arellano and Bover, 1995) transform the data by subtracting the mean of future observations rather than the prior observation. The transformation $x_{it}^* = x_{it} - \bar{x}_{i,t+1:T}$ uses only the focal observation and the average of forward observations, so a gap at t does not eliminate $t - 1$. Second, system GMM (Blundell and Bond, 1998) adds a levels equation instrumented by lagged differences.

Fig. 2 compares four estimator variants. At $K = 20\%$, difference GMM with first-differences retains 64 percent of baseline observations. All three alternatives retain approximately 80 percent. System GMM with orthogonal deviations retains no more observations than system GMM with first-differences; the levels equation is the binding constraint on sample size.

Note that orthogonal deviations recovers observations through its regressor transformation, not through the instrument set. The benefits of these alternatives extend beyond observation count. Under HENR zero-substitution, surviving observations are retained but their effective instrument matrix has zeros where complete data would have informative lags, weakening identification of coefficients on lag-instrumented regressors. Fig. 3 reports the degradation factor, defined as the empirical growth rate of $SE(\hat{\alpha})$ relative to the $1/\sqrt{N}$ growth rate that observation loss alone would predict. A factor near one means observation count fully explains efficiency loss; a factor above one means surviving observations have weaker effective instruments. At $K = 20\%$, difference GMM with first-differences shows a degradation factor of 1.51, while orthogonal deviations stays close to 1.05. Combining observation count and instrument strength, the effective sample for inference under difference GMM with first-differences is roughly 28 percent of baseline at $K = 20\%$, compared to 72 percent for orthogonal deviations.

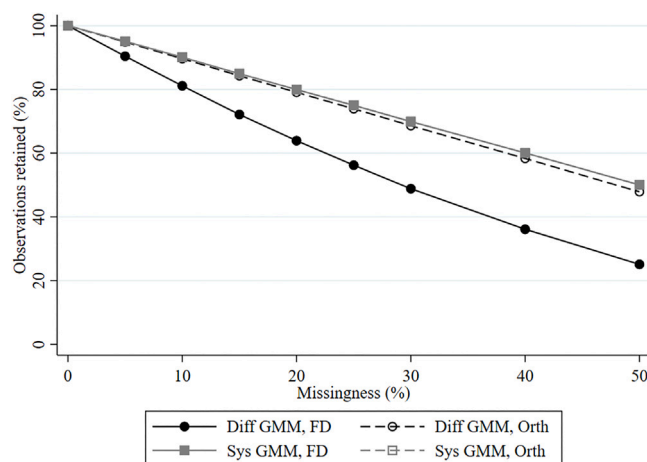


Fig. 2. Observation retention by GMM variant. Note: System GMM with first-differences and system GMM with orthogonal deviations overlap completely.

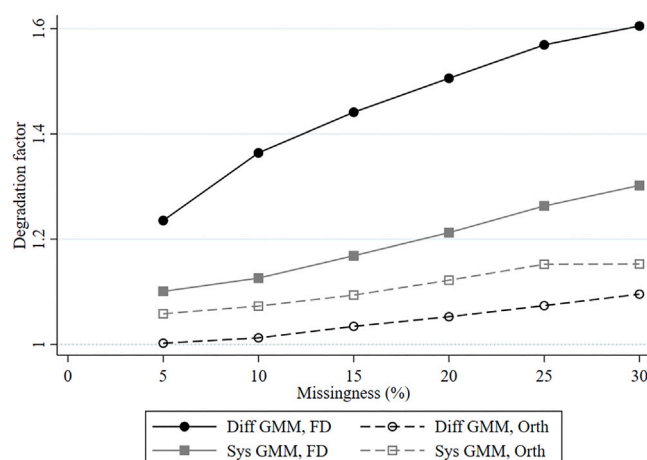


Fig. 3. Degradation factor for SE of the lagged dependent variable, by GMM variant.

4. Discussion and conclusion

Under the MCAR assumption used in these simulations, the cascade affects only efficiency. Surviving observations remain a random subsample, so GMM consistency is preserved; standard errors are larger, but coefficient estimates are not biased. Real-world missingness is rarely random, however, and non-random missingness can introduce sample selection. Firms with complete data may systematically differ from firms with gaps, and because the cascade clusters observation loss around each missing value, any such selection is amplified. These selection concerns arise from the missingness mechanism, not from GMM mechanics.

Several patterns of non-random missingness are common in finance datasets. Small firms report less data than large firms; distressed firms suspend disclosure of selected items; and certain line items such as R&D, segment data, and governance variables were sparsely disclosed in earlier decades and now appear with substantial historical gaps. Quantifying observation loss under MCAR nevertheless provides a useful benchmark, since even this optimistic scenario reveals substantial and previously undocumented attrition.

These findings have several practical implications for finance research. First, reported GMM sample sizes reflect substantial attrition beyond raw missingness. A study with 5 percent missingness on one endogenous regressor may have lost 10 percent of potential observations to the cascade, and the loss is much larger for variables such as R&D or advertising where missingness can exceed 50 percent. At a 50 percent missingness rate, difference GMM with first-differences retains only one quarter of baseline observations even with a single endogenous regressor.

Second, the compounding effect with multiple endogenous variables suggests a sample size argument for parsimony. Adding a third endogenous regressor with 15 percent missingness can reduce the usable sample by more than 60 percent, which may outweigh

any gains from additional controls. In capital structure or governance applications where investment, leverage, and governance variables are routinely treated as jointly endogenous, this consideration is binding.

Third, orthogonal deviations offer a simple remedy. The `orthogonal` option in `xtabond2` implements this transformation at no cost to identification. The simulations here suggest that orthogonal deviations can increase usable observations by 25 percent or more and additionally preserve the strength of surviving instruments.

Finally, imputation is particularly valuable in GMM contexts because the cascade amplifies each imputed cell into multiple recovered observations. With three endogenous variables, reducing missingness from 20 percent to 10 percent more than doubles the usable sample, from 26 percent to 53 percent.

Declaration of generative AI use

During the preparation of this work the author used Claude for coding and debugging, as well as for checking grammar. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Antoniou, A., Guney, Y., Paudyal, K., 2008. The determinants of capital structure: capital market-oriented versus bank-oriented institutions. *J. Financ. Quant. Anal.* 43 (1), 59–92.
- Arellano, M., Bond, S., 1991. Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. *Rev. Econ. Stud.* 58 (2), 277–297.
- Arellano, M., Bover, O., 1995. Another look at the instrumental variable estimation of error-components models. *J. Econometrics* 68 (1), 29–51.
- Beck, T., Levine, R., Loayza, N., 2000. Finance and the sources of growth. *J. Financ. Econ.* 58 (1–2), 261–300.
- Blundell, R., Bond, S., 1998. Initial conditions and moment restrictions in dynamic panel data models. *J. Econometrics* 87 (1), 115–143.
- Bun, M.J., Windmeijer, F., 2010. The weak instrument problem of the system GMM estimator in dynamic panel data models. *Econom. J.* 13 (1), 95–126.
- Coën, A., Racicot, F.-É., 2007. Capital asset pricing models revisited: Evidence from errors in variables. *Econom. Lett.* 95 (3), 443–450.
- Flannery, M.J., Hankins, K.W., 2013. Estimating dynamic panel models in corporate finance. *J. Corp. Financ.* 19, 1–19.
- Hayakawa, K., 2007. Small sample bias properties of the system GMM estimator in dynamic panel data models. *Econom. Lett.* 95 (1), 32–38.
- Holtz-Eakin, D., Newey, W., Rosen, H.S., 1988. Estimating vector autoregressions with panel data. *Econometrica* 1371–1395.
- Koh, P.-S., Reeb, D.M., 2015. Missing R&D. *J. Account. Econ.* 60 (1), 73–94.
- Levendis, J., 2025. Default and collapsed instruments in dynamic panel GMM: A Monte Carlo comparison. *Econ. Bull.* 45 (4), 1915–1925.
- Racicot, F.-É., 2015. Engineering robust instruments for GMM estimation of panel data regression models with errors in variables: A note. *Appl. Econ.* 47 (10), 981–989.
- Racicot, F.-É., Rentz, W.F., Théoret, R., 2025. Is illiquidity priced in an international factor pricing model? A dynamic panel data application with robust IV. *Int. J. Financ. Econ.* 30 (1), 282–314.
- Roodman, D., 2009a. How to do `xtabond2`: An introduction to difference and system GMM in stata. *Stata J.* 9 (1), 86–136.
- Roodman, D., 2009b. A note on the theme of too many instruments. *Oxf. Bull. Econ. Stat.* 71 (1), 135–158.
- Windmeijer, F., 2005. A finite sample correction for the variance of linear efficient two-step GMM estimators. *J. Econometrics* 126 (1), 25–51.
- Wintoki, M.B., Linck, J.S., Netter, J.M., 2012. Endogeneity and the dynamics of internal corporate governance. *J. Financ. Econ.* 105 (3), 581–606.